

McGill Baja Racing - October 3-5, 2025 - Seven Springs, NY

(Note from last time: the car's name turned out to be Murphy.)

Friends, foes, and miscellaneous,

Hello, and welcome back to another McGill Baja Racing newsletter! In addition to friends and family, our last newsletter had another surprise audience: sponsors! (The team has multiple sponsors, who donate money, materials, and services. Without them, the car would not be possible.) To the sponsors reading this: hi! Thanks for supporting us. <3



Our need for speed is only superseded by our need for sponsors.

This newsletter covers our latest competition, which was an unofficial competition in upstate New York called Oktobajafest, hosted by Clarkson University. As with before, explanations for non-technical readers are in (blue) and more technical details are in [red]. (Official competitions are planned by the Baja SAE (Society of Automotive Engineers) organization and have very strict procedures. Unofficial competitions are entirely organized by the host university, and they can modify things to their liking. We like both!)



Up,

Whether attending official or unofficial competitions, McGill Baja Racing does do one thing consistently, and that is packing at the last minute. This time we had a good reason (we always do!), which was that the bevel casing was getting machined at the last minute. [Last comp, 4WD stopped working midway through endurance. After opening the case up, we found out that this was caused by a bearing breaking in the bevel casing. Getting it re-designed turned out to be a feat and a half, as there was already no extra space to put in a stronger bearing. To fix this, we ended up extending the shaft (and the casing) downwards and turning the gearbox into what resembled a miniature Frankensteined labyrinth (happy Halloween!). It took awhile to troubleshoot and machine, so much so that the machining ended the afternoon of packing. The Drivetrain team put it in, made sure that everything spun to some degree, and called it a day.] Packing started at 1pm, and ended at around 1am. The truck left at 5am the next day. Talk about a quick turnaround.



And over!

Day 1: Arrival & Tech

The drive down to Clarkson takes around 2.5 hours, so while the truck left early to get a good trailer spot at the competition, the rest of us left at around 6am. The drive was beautiful—the sun rose on autumn multicoloured trees as we passed picturesque houses and long fields of corn. It was great from an engineering standpoint too, as we got to see some cool tractors.

([Video](#), slide 1: Do you think we have a future in content creation?)

Seven Springs, property of Clarkson University, is a strip of former forest that was cleared into a ski hill, according to a sprightly elderly local wearing a wide-brimmed beige fabric hat. It is now used as a dirt bike track, plus for the annual Oktobajafest. The setting is rolling hills steeped with red and orange fall colours, a sparkling blue creek, and large friendly dogs being walked. Weather-wise: three days of sun, no humidity, and temperatures of brisk fall mornings, warming up in the afternoons. In other words, one couldn't ask for more. Unfortunately, it seems that this weather had lasted for several weeks prior, and the amount of dust that even walking kicked up made a scene that looked straight out of a Western.



The beautiful Oktobajafest grounds. You can just see the porta-potties to the right.

Oktobajafest lasted three days. Day 1 was Tech (quick reminder: Tech stands for Technical Inspection and involves making sure that the car build adheres to all the safety rules). To encourage us to pass Tech properly and efficiently, the organizers let us know that there was a 50-point bonus for whichever team passes Tech with the least number of alterations. ("Tape and zip ties are fine. The minute you have to get a wrench out, go back.") After triple-checking our car, we sent it over to the Tech judges. We were sure we had a shot at that prize! We did not. We ended up being the second-to-last car to succeed. The culprit? Brakes as usual. I won't bore you with the details. If you know them, you don't need to hear them again. If you don't, stay in blissful ignorance.



Captain Jemma sitting sadly on a pile of gravel. Cheer up, Jemma! It'll be over soon.

[And by brakes, I mean the static brake check, not the dynamic one. The car straight up rolled when pushed. On the other hand, because we worked on the brakes for so damn long during Tech, the actual Brake Check went swimmingly.]



Brake fluid is a key element to a well-rounded skincare routine.

Being engineering students, we are car nerds second, and student nerds first (unfortunately). A selection of topics was studied by our team members, notably: controls, materials, biomaterials, dynamics, machine element design, heat transfer, deformable solids, and aerospace structures. Driver Alvaro would like to add that he studied nothing and regrets it. He probably also regrets the fact that he hit a tree during our practice run, but that's on Day 2. (Catch strays, Alvaro!)



Suspension Lead Isaac horizontal, part 1.

Day 2: Dynamic Events

After a night spent either studying or eating takeout from "A1 Oriental Kitchen" and lounging at the only working hotel pool in Baja competition memory, Saturday consisted of doing a practice run of the car (as we hadn't actually tested to see if our repaired drivetrain actually worked). Pro: the car ran! Con: the practice track was in the woods, which are famously filled with trees. While doing a routine run, Alvaro hit a rock, spun, and hit a tree. [Bonk.] Luckily, only a small suspension bracket had to be replaced. Unluckily, it turned out that out of the two-part bracket assembly, four of the top part were packed, and only one of the bottom part. This meant that if we replaced the bracket now, before the start of even the dynamic events, we had no replacement for when competing. The solution devised had the following method, completed by

Suspension Lead Max: put the vise onto the trailer ramp, draw the desired bracket shape onto 1/4" sheet metal with a sharpie, clamp the sheet metal vertically into the vise, start up the generator, take an extension cord and plug one end into the generator and the other end into the angle grinder, and freehand angle grind out the bracket. (To any professors reading this: safety glasses were worn.)

([Video](#), slide 2: A unique maneuver!)



Not sure if this is how they do things professionally. Thoughts?

[The broken bracket is pictured below. Note that the top and bottom parts are welded; this is because the cascading failure system in fact worked too well, and the bracket bent too many times. We therefore welded it into a single piece for strength, and it's quite good for bottom impacts. Unfortunately for the entire endeavour, this was a front impact.]



New addition to the Wall of Shame! ([which is a real thing.](#))

On to the actual events. As it's an unofficial competition, the event names changed slightly. Maneuverability was named "Short Track," and S&T was "Baja Cross." (Hill Climb remained Hill Climb.) (Reminder: dynamic events are the short events that take place the day before Endurance, and test the car's steering and power capabilities. S&T stands for Suspension & Traction, i.e. can it go on rocks and jumps and not get stuck. Cars have two attempts to complete each event.) Here are the results:

- Short Track: not great. On the first run, the driver accidentally hit the kill switch. On one hand, whoops. On the other hand, having a kill switch that is too easily accessible is not the worst problem to have (see last newsletter when we were nearly run over due to a neighbouring runaway car with a nigh-inaccessible kill switch). On the second run, a nut came loose and a suspension member [lower A-arm] fell off. Whoops!!



We look so fast here! We were not.

- Hill Climb: decent. We made it up both times, but other cars got up faster. [The first try was 4WD and the second was 2WD. We were faster with 2WD.] The vehicular hill climb was shortly followed by a human hill roll.



Driver and Brakes Lead Nadim performing the Hill Climb.



Zac and Tatiana performing the optional second, lesser-known part of Hill Climb.

- Baja Cross: trials, tribulations, and triumph.

The Baja cross track was a short way away from the main competition area. In the woods, between trees, the track was a path is one of ramps, boulders, and steep turns. In key points were hay bales which could be easily moved to expose exit paths, which were used by car after car as each in turn got stuck and had to bail out. The team was apprehensive. Would we make it? Would the car stay intact? Driver Remy was not concerned. Quote: "Imma send it. I don't care about the car. You guys are not sleeping." ([That last sentence alludes to all-night repairs, which, while sometimes necessary, are not exactly welcomed.](#)) Remy is a man of his word:

([Video](#), slide 3: The wheels on the baja go round and round! Credit: Clarkson University)

Luckily for Remy and all others involved, the rollover produced only minor damage, and after a few minutes of repairs, we were ready to go again. **[The brake electronics disconnected, and the front belly panel needed a patch.]** The second run was much more successful. Remy cleared obstacle after obstacle. Even the giant tractor tires, which had proved to be his original downfall, were no match. He neared the final obstacle: two steep ramps, one after the other. He made it up the first one, but the speed decrease stalled him out at the second. He backed up and tried again, with more speed. Stuck again. You could see him frantically switching between 2WD and 4WD, wiggling the steering wheel, practically willing the car to move. We held our breaths. Come on, Remy. You can do it. Remy backed up to the top of the first ramp, and gunned it down the slope, up the next, and over, and a quick slalom, and he had done it! Yells of celebration across the course. Cheers from the other spectators. We were one of the only two cars out of 15 that had completed the Baja Cross track! Murphy's name stands true: anything that can happen will happen.

([Video](#), slide 4: Hooray!)

Other miscellaneous events: Brakes Lead Nadim organized an impromptu birdwatching outing, "fun and interesting" social media content was created in the absence of Communications Director Manon (strangely, she did not approve), and the porta-potties were overrun by invasive Japanese lady beetles, which got into ears and other places.



Nadim surrounded by nature, after a day surrounded by brake fluid.

That evening, a few others and I left early. "Go back to the hotel," Captain Jemma said. "We're just going to tweak a few things; we'll be right behind you." Famous last words. Captain Lucas has helpfully provided a description of the night's activities—key parts are highlighted by yours truly.

4pm dynamic events are done, huge success for the team, getting ready to pack up and **head home for an early night** for once after a few suspension adjustments and some brake tuning, I think. Half of the team headed home around 4:30, some people did endurance track walk and driver's meeting, and some were adjusting/reinstalling suspension. We noticed some **clicking in the diff** and wanted to determine the source of it. At around 6pm suspension was finally done reinstalling, and we were able to run the car on sawhorses to listen closely for any diff noises or any other irregularities in the system. Some noise was heard in the diff but nothing out of the ordinary. I then heard more knocking coming from the rear at the center driveline linkage. Feeling on the guarding I could tell something was knocking against it which isn't supposed to happen. We took off the rear guarding and found that one of the caps on the u-joint cross had come off of the double-cardan joint in the rear. Not only that but the tiny bolts holding the bearing carrier to the mounts were backed out, **allowing the whole system to flail around** and be unconstrained. This loosening must've caused the cap to come off of the cross. This was already bad, but we thought we could repress a cap onto the joint while it was still on the car to avoid removing all the retaining rings and annoying linkage back there. We tried to do so using a clamp positioned as in line as possible to hopefully press the cap right back on. Unfortunately, after clamping very hard and even using steering's lower column spare as a breaker bar to gain more leverage (sorry steering, **the column is now bent and unusable!**), we were **unsuccessful** in pressing the cap back on. I then spent a while **under the car with oil leaking close to my face** removing the linkage and retaining rings to free the double cardan joint so that we could work on it with the proper press. The tiny bolts on the bearing carrier were located and tightened back in place in the meantime. With the joint out of the car, we saw that the cap had gone in at an angle and thus was **lodged in the double cardan joint**. While removing the angled cap we **bent** the ears of the u-joint, which is **catastrophic**. By this point it is already close to 8pm and we were **hoping this would be a quick fix**. We then **spent the next two hours** attempting to unbend the ears of the cardan joint in order to repress a cap in the correct way. Zac used a **combination of socket heads and random sized fasteners** to strategically try and pry the ears back open by wedging items between them and using **all of Ben Guan's body weight** on the press to do so. At this point the only people remaining are me, Ethan, Zac, Julian, and Ben. While we were able to straighten the ears, we **jammed** the joint even more in the process and **almost buckled the whole cross** with all the weight we were applying to the piece, so the original joint was **deemed unfixable** at around midnight. By this point, Jemma, Remy, and Aidan arrived back at the site to bring us **pizza** which was very nice. We brought out the spare double joint and began to assemble it entirely, with all 8 u joint caps to press on. This did take a while but went smoothly as the piece was straight and the caps should press on easily if everything is aligned. At around **2am** we **finally** finished reinstalling back on the car and the car was **ready to roll** out for endurance the next morning.



Red baja at night, sailor's delight? No, that's not it...

Day 3: Endurance

Woohoo, main event day! We arrived at 7am, full of fresh country air and sleep deprivation. (At our arrival, we witnessed, in awe, a large man from the Clarkson team singlehandedly pick up an entire couch out of their trailer and casually carry it out.) The morning was spent putting some last tweaks on the car and getting the paddock (the area next to our trailer where the car gets fixed) ready for the race.



Suspension Lead Isaac horizontal, part 2.

The day advanced. Gridding (cars getting in prescribed order for the race) started, engines turned on. The custom CVT squealed a little. (It's ok to express nerves before a big event!) (Reminder: CVT stands for continuously variable transmission and is the baja car's version of gear changing. It's composed of a belt and two pulleys that change sizes thanks to springs and weights.) The cars slowly wound around the grounds in a long procession. Golden morning light

shone through the trees as they disappeared into the forest, silent but for the engine sounds and squeaky CVTs. (The standardized Baja engine, used by all teams, is that of a lawnmower.) And then it started and there they were, back into view, roaring up the first hill (slightly slowly, due to the low power of lawnmower engines when going uphill), disappearing once again in a cloud of dust into the trees.



Nyoom!

Murphy started 10th, due to the not-too-fast performance during Hill Climb. However, she soon started gaining. A few laps in, and we were up to 6th. Win: we got all the way up to 2nd place. Loss: the CVT broke, and we had to get towed in. Win: the repair only took 15 minutes. Loss: we were back on the track down to 5th. Win: we came back up to 2nd, which is where we finished! Quote from Composites Lead and Driver Communicator Aidan, over the radio: "Big dubsbies. Over."



Strike a pose!

[This comp, we ran a custom primary and secondary. The secondary was newly machined, but parts of the primary were reused from prior years. We had a small fatigue failure in the old part and could have reassembled it, but chose instead to completely replace it with the CVTech CVT to save time. We kept the custom secondary in.]



Transmission Lead Brett replacing his baby.

In total, we won two plaques: one for our 2nd in Baja Cross, and one for our 2nd in Endurance. [The points aren't weighted towards Endurance, so we didn't get any overall prizes.] Thanks to Oktobajafest's numerous sponsors, we also got two angle grinders as prizes—we traded one of them for an impact drill with the University of Western Ontario. (This was the first-place prize, and they got four of them, plus the overall 1st place trophy. Congratulations, Western!)

([Video](#), slide 5: Keeping Up With The McGillians – taken post-comp)

On our way home we took a break on a Quebec farm to watch the sunset. We let the farmer's kid take a look at Murphy.



Is this how MacDonald Campus feels?

And that's it! No near-death experiences this time, but a great event nonetheless.



The team. Love you all.

Hugs, kisses, and fist bumps,
Ada Santavy



Author pictured left, teaching Captain Jemma how to shadowbox.

Thank you for getting to the end of this newsletter. If this was forwarded to you and you would like to subscribe for more updates from McGill Baja Racing, click [here](#). If you have feedback for us or no longer want to receive these messages, please reply (do not Reply All!) to this email.